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ARTIFICIAL INTELLIGENCE FOR ROBOTICS 1

Group 8 Assignment

Autonomous tractors for plowing, planting and irrigating

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1 Introduction

An **autonomous tractor** is a *driver-less* agricultural vehicle that combines sensors, artificial intelligence (AI), and drones to operate without direct human intervention [1]. These technologies enable the tractor to navigate the environment, perceive relevant conditions, and make decisions autonomously in order to perform farming operations efficiently.

In this project, the autonomous tractor operates within a previously mapped agricultural area. The land is divided into different areas according to soil characteristics, allowing the system to associate each area with suitable crops and its correct type of irrigation system. This structure supports the decision-making processes related to **soil preparation**, **crop sowing**, and **irrigation**.

Project Assumptions

This project assumes that the agricultural field is a controlled and secure environment, where access by humans or animals is not allowed. As a consequence, obstacle detection and avoidance are not considered within the scope of this work. This assumption allows the focus to remain on the agricultural tasks performed by autonomous tractors.

To further simplify the case, multiple autonomous tractors are assumed to be specialized in a single operation, so they are equipped with a single tool. The tasks considered in this project are soil preparation, sowing, and irrigation. In order to accomplish these tasks, the autonomous system must be able to recognize fields, identify soil types, classify crops, and determine the appropriate irrigation methods based on crop requirements and field conditions.

Plants that grow on trellises were also excluded from this project. The crops considered are *open field crops* that grow directly on the soil. This assumption avoids cases in which the tractor would be unable to move, and it also excludes the existence of greenhouses, which would require additional sensors and human supervision.

Method and Goal

The reasoning model adopted in this project is based on an **ontology**, which provides a structured and formal representation of domain knowledge. An ontology models agronomic concepts as classes, relationships, and data properties. Using the **Web Ontology Language (OWL)**, it can support semantic reasoning and guide the behavior of autonomous systems [2].

The main goal of this project is the development of an ontology that supports autonomous tractors during plowing, planting, and irrigation operations. The ontology is designed to represent the essential entities and relationships of the agricultural domain in a clear and task-oriented manner, allowing the robotic system to reason about crops, fields, soil properties, and irrigation strategies.

2 Domain Overview

The considered project involves multiple areas, each divided into specific fields based on the properties of the soil. The focus of this project is the analysis of farming activities, such that the tractor can autonomously decide the most appropriate task to perform.

Each agricultural operation is performed through a specific tool mounted on the autonomous tractor which directly constrain the actions that can be executed, this action are **soil preparation**, **sowing** and **irrigation**.

The scope is to give the tractor the ability to identify fields and crops, understand soil conditions, and select appropriate actions based on requirements and seasonal constraints.

3 Modelling choices

The ontology is centered around the *AutonomousTractor* class, which is divided into subclasses depending on the vehicle power.

The tractor needs to reason about three possible *AgriculturalOperation*. For this purpose, the *Tool* class was introduced and divided into specific tools for each action, with additional subclasses representing different tool types. This additional division is necessary for each type of *SoilType*. The *SoilType* class models different soil categories in order to represent the possible environments the robot may encounter although, only a subset of them is used in the real agricultural fields. Each soil type is also linked to a qualitative *PhLevel*, which allows the selection of the most appropriate crops based on soil characteristics and pH requirements. Crops are associated with specific *Field* instances, since multiple compatible crops can be planted within the same area. Therefore it is possible to map the farm structure.

Additionally, *Season* was modeled through individuals rather than classes, allowing the ontology to represent seasonal preferences of crops and to improve the crop–field mapping.

To complete the assignment, a reference farm in which all instances of the modeled entities exist was considered. In this context, *Object Properties* are interpreted as existential relationships, meaning that the existence of a relation represents compatibility within a realistic agricultural setting.

Following, an example of the main relationship used in the project:

Domain	Property	Range	Description
AutonomousTractor	hasTool	Tool	Specifies the tool equipped by the autonomous tractor
Crop	needsTypePlowing	SoilPrepTool	Specifies the soil preparation tool required by a crop
Crop	needsTypeSeeding	PlantingTool	Indicates the planting tool suitable for sowing the crop
Crop	needsTypeIrrigation	IrrigationTool	Defines the irrigation tool required by the crop
Crop	needsPhLevel	PhLevel	Represents the preferred soil pH level for the crop
Crop	seasonPreferred	Season	Indicates the preferred growing season for the crop
Crop	isPlantedIn	Field	Associates a crop with the field where it is planted
Field	isSoilTypeOf	SoilType	Defines the soil type associated with a field
SoilType	hasPhLevel	PhLevel	Links a soil type to its characteristic pH level
Tool	performs	AgriculturalOperation	Indicates the agricultural operation performed by a tool
UAV	hasSensor	Sensor	Associates a UAV with its onboard sensors
Crop	canBePlantedWith	Crop	Represents crop compatibility for co-planting (symmetric relation)

Table 1: Main object properties used in the ontology

4 Domain Analysis

Agricultural Operation

The *AutonomousTractor* implemented subclasses to distinguish the tractor by power, this is necessary to conjoin the Tractor to the *AgriculturalOperation* and the corresponding *Tool*[3].

According to classical agricultural mechanization literature, tractor classification, into low, medium, and high-power categories, is based on the power required by the field operation [4]. For this project scope, only the low-power and high-power are used.

Operation	Purpose	Power Requirement
Soil preparation	Prepare terrain for sowing	High Power Vehicle
Sowing	Plant seeds	Low Power Vehicle
Irrigation	Water supply	Low Power Vehicle

Table 2: Agricultural Operations and required Power Vehicle

Tools

Each autonomous tractor can use only one tool at a time (the property *hasTool* is functional). Different crops need different tools for the same operation [5]

- **Soil preparation tools**[6]

These tools mechanically manipulate the soil profile to establish a viable seedbed **DiffHarrowPlow**.

- **Plow**: designed to invert the upper soil layer. Its function is to bury crop residues and weeds while aerating the substrate to facilitate root penetration. It is used for deep tillage.
- **Harrow**: used for surface refinement. It pulverizes soil clods, levels the field topography, and mechanically disrupts emerging weed growth. It is used for shallow tillage.

- **Seeding tools**

These tools perform the spatial distribution and deposition of seeds into the soil matrix.

- **Air seeder**: A pneumatic system that uses a high-velocity air stream to convey seeds from a central hopper to the soil. It is optimized for high-efficiency planting across large acreages.
- **Seed drill**: A mechanical implement that places seeds in continuous rows at a controlled depth and covers them.
- **Precision planter**: A sophisticated sowing instrument engineered to separate seeds and place them at exact intervals and depths. This is critical for crops requiring specific plant-to-plant spacing.
- **Transplanter**[7]: An agricultural machine designed to transplant seedlings from nurseries to the field.
- **Potato planter**: A specific tool to plant potatoes.

- **Irrigation tools**¹

These tools facilitate the artificial application of water to agricultural lands.

- **Boom sprayer**: A horizontal structure equipped with multiple nozzles. It delivers water in a uniform way over the crop [8].
- **Drip line deployer** : A tool used to install drip irrigation tubes that enable micro-irrigation directly to the root zone. This operation needs to be done only once after the sowing.

Sensor

In the transition from traditional farming to smart farming, the role of sensors is central. In order for the system to operate effectively, it must continuously acquire accurate data about the environment.

To increase the precision of data acquisition, two types of sensors are implemented:

¹Irrigation operations are complementary and compatible. If two crops are planted in the same field and require two different types of irrigation, both methods are performed in said field.

- **StationarySensors**[9]

Fixed in the field (object property *hostBy*)

- **HumiditySensor**: essential for monitoring atmospheric and soil conditions that directly influence plant growth rates and water requirements.
- **TemperatureSensor**
- **PhSensor**: crucial for determining whether a field's soil acidity level is suitable for crops with different pH preferences.
- **WindSensor**: improves operational safety and efficiency, particularly by identifying favourable conditions for spraying activities or aerial operations involving unmanned aerial vehicles (*UAV*).

- **UAVSensor**

Sensors mounted on an UAV (object property *hasSensor*)

- **LIDAR**: supports topographic mapping and biomass estimation, while the *SAR* enables soil moisture monitoring and all-weather mapping [10].
- **RGBCamera**: used for crop growth monitoring and disease detection.
- **SAR**: (Synthetic Aperture Radar) emits microwaves to generate high-resolution images of the terrain and operates with no regards to weather and moment of the day.

All sensors transmit data in real time to the autonomous tractor through the network (property *sendsData*).

The integration of unmanned aerial vehicles (*UAV*) enhances efficiency, sustainability, and data-driven decision making and enable real-time crop health monitoring and optimized resource management, leading to increased productivity with significantly reduced resource consumption.

SoilType

The type of soil is fundamental to determine which crops can successfully grow and thrive, and it is necessary to determine the agricultural operations required for cultivation. The ontology models 10 distinct **soil types** [11] [12]: *Calcareous*, *Clay*, *ClaySandy*, *Gravel*, *Loamy*, *Organic*, *Peaty*, *Rocky*, *Sandy* and *Silty*.

They differ from each other in terms of chemical composition, soil texture and structure, and pH values.

The chemical composition considers the presence of organic matter and carbonates, it also refers to the capacity of the soil to hold nutrients, therefore this property influences fertility and crop suitability. Fertility depends on the interaction between organic carbon, nutrient availability, soil structure, and management practices [13]. For this reason, the model implements *Organic* soil type as a descriptive category indicating high organic matter content, rather than as a direct indicator of soil fertility.

The texture of the soil refers to particle size distribution and influences the required agricultural operations.

Soils with larger particles drain more easily, while finer-grained soils retain water and nutrients but are more prone to compaction.

Each *SoilType* is connected to a *PhLevel* through the existential object property *hasPhLevel*, as shown below.

PH Level	Soil Type
Acidic ($pH < 5.5$)	Sandy, Peaty, Organic
Neutral ($5.5 < pH < 6.5$)	Loamy, Gravel, Clay, Sandy, Silty, Organic, Rocky, Clay Sandy
Alkaline ($pH > 6.5$)	Calcareous, Clay

Table 3: Qualitative pH levels associated with soil types

The ontology implemented all soils to give the Tractor a structured model so it could recognize the type of soils, however only three are really used:

- **Clay**

it is composed of fine-grained particles, retains nutrients well, but drains poorly, making it difficult to work when wet. Typically has alkaline to neutral pH.

- **Loamy**

it is composed of sand, clay and silty in equal quantities, therefore shows intermediate characteristics and usually has a neutral pH.

- **Sandy**

it is composed of large-grain, it drains quickly and tend to have acid to neutral pH.

This representation allows the autonomous system to reason about soil conditions and to select compatible crops and appropriate agricultural operations for each field.

Crop

The ontology considers 17 different vegetable crops divided by their agricultural family (*Apiaceae*, *Brassicaceae*, *Cucurbitaceae*, *Fabaceae*, *Poaceae* and *Solanaceae*).

Each crop is linked to the preferred *PhLevel* to grow, the compatibility with other crops, the operation they require for planting and growing and the preferred season.

The preferred soil acidity level is modeled through the object property *needsPhLevel*. Rather than using numerical pH values, the ontology adopts a qualitative classification into *Acidic*, *Neutral*, and *Alkaline*, which is sufficient to capture compatibility [14]. This choice allows the autonomous system to reason about crop suitability based on soil characteristics and to exclude incompatible planting options.

To indicate the compatibility relationship between crops [15] [16], the existential restriction *canBePlantedWith* that means "There exists a crop *X* planted near a crop *Y*" was implemented, and it was used to demonstrate the compatibility between the crops *X* and *Y* and it is considered as symmetric property.

Examples of asserted compatibility pairs include:

- **Cabbage** *canBePlantedWith* Beans, Celery, Cucumber and Potatoes;
- **Cucumber** *canBePlantedWith* Beans, Broccoli, Cabbage, Cauliflower, SweetCorn, Peas and Radish;
- **Eggplant** *canBePlantedWith* Beans, Peas and Potatoes.

It is possible to see the **Protégé** implementation in the following image:

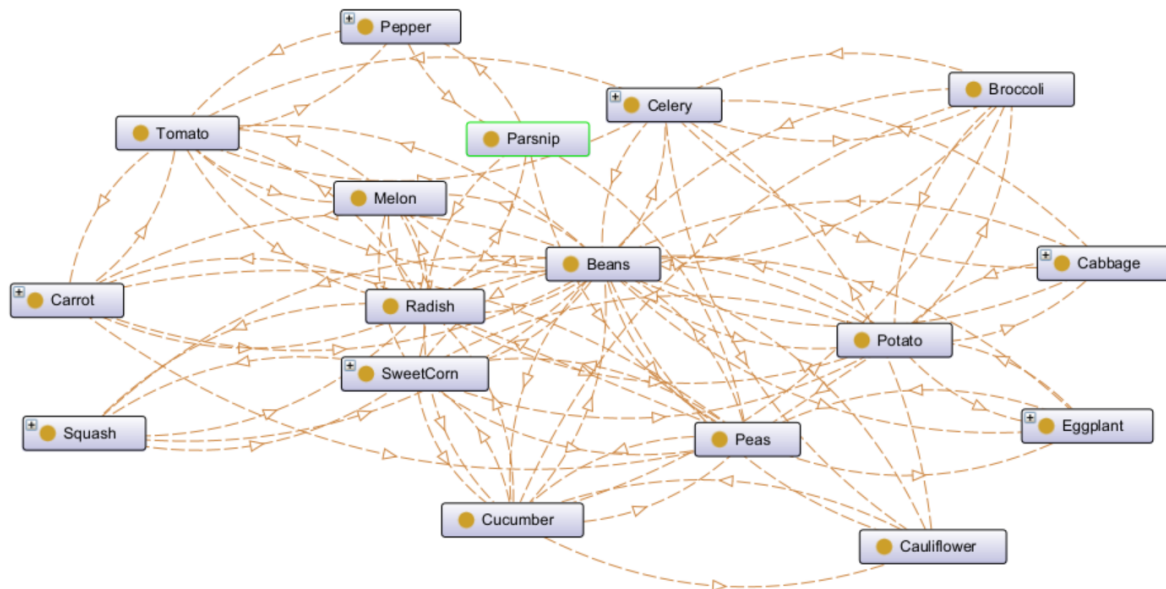


Figure 1: Graphical representation of the *canBePlantedWith* property

This modeling choice enables the relationship between crops and vehicles: crops specify *what* operations are needed, while tractors determine *how* these operations are executed, based on the tools they are equipped with. An example of the *AgriculturalOperation* required by different crops[5] is reported in **Table 4**.

Crop	Soil Preparation	Sowing	Irrigation
Carrot	Harrow, Plow	SeedDrill	DripLineDeployer
Radish	Harrow	SeedDrill	BoomSprayer
Potato	Plow	PotatoPlanter	BoomSprayer or DripLineDeployer
Tomato	Harrow	Transplanter	DripLineDeployer

Table 4: Example of crop requirements and related operations

The preferred planting season, implemented as existence relationship as **seasonPreferred** (shown in **Table 5**) enables the autonomous system to plan planting schedules and crop rotations. The season analysis refers to Mediterranean climate[17].

Season	Preferred Crops
Winter	Celery, Parsnip, Squash
Spring	Broccoli, Peas, Potato
Autumn	Cabbage, Cauliflower, Beans
Summer	Carrot, Radish, Cucumber, Melon, SweetCorn, Eggplant, Pepper, Tomato

Table 5: Season preferences

By combining crop requirements with soil conditions, seasonal constraints, and agricultural operations, the ontology enables informed decision-making for autonomous cultivation planning.

Fields

Based on the evaluations performed on crop compatibility, soil characteristics, and seasonal constraints, the farm was subdivided into a minimal set of fields ensuring feasible cultivation conditions[18]. Therefore, the farm was divided into 8 areas, each containing at least one crop. The number 8 was chosen as the minimum number of areas to plant all the crops, and each area containing the highest number of compatible crops, taking also into account the *SoilType* compatibility².

The following table lists all the field combinations within the farm:

Area	Soil Type	Crops combination
1	Sandy	Parsnip, Pepper
2	Sandy	Beans, Cabbage, Celery
3	Sandy	Eggplant, Peas, Potato
4	Loamy	Squash, SweetCorn
5	Loamy	Broccoli, Cucumber
6	Loamy	Cauliflower
7	Clay Sandy	Carrot, Radish, Tomato
8	Clay Sandy	Melon

Table 6: Fields distinction

In this table, *Sandy* soil is considered as *Acidic* for modeling purposes, in order to ensure a more balanced distribution of soil conditions across the fields.

Furthermore, each *Area* is modeled as an *individual*, enabling explicit assertions about which crops are planted where and supporting consistency checks on compatibility and soil constraints.

The following image provides a schematic representation of a possible farm layout, showing the subdivision into areas and the associated crop combinations:

²In case of combinations having more than one preferred *SoilType*, one of them was arbitrarily chosen.

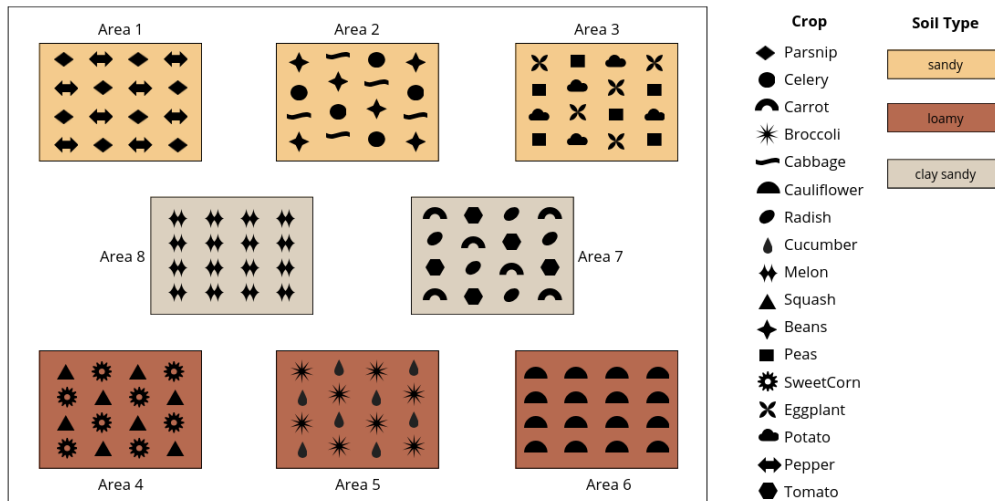


Figure 2: Example of field division

This field subdivision demonstrates how crop compatibility and soil constraints can be combined to generate a feasible farm layout.³

5 Ontology Testing and Validation

The ontology was validated using standard ontology engineering practices[20]. The validation process focused on verifying logical consistency, coherence of class hierarchies, and reasoning capabilities.

The ontology was implemented and tested using *Protégé*, exploiting OWL-DL expressiveness and automatic reasoning. A reasoner was employed to check the absence of logical inconsistencies, ensuring that no contradictory class definitions or unsatisfiable concepts were introduced.

The testing process is divided into two distinct phases:

1. Introducing deliberate modifications to the data to assess whether the reasoner correctly detects inconsistencies

A primary focus of this phase is validating constraints regarding crop incompatibility within the same cultivation area. To perform this test effectively, individuals must be instantiated for each crop class; without these instances, the reasoner may treat empty classes as valid, failing to trigger errors even in logically invalid scenarios. For example, when an instance of Celery and an instance of Parsnip are asserted within the same area, the reasoner reports an inconsistency. This behavior validates the ontology, confirming that the constraint preventing these two crops from coexisting is functioning correctly.

2. Using the Protégé tool DLQuery to extract results

Testing with Description Logic (DL) queries is a fundamental method for verifying and validating the logic structure of an ontology. The process relies on the semantic reasoner *Pellet*. Queries were defined using class expressions and object property restrictions to retrieve inferred knowledge rather than explicitly asserted facts [21].

Therefore, the validation process is based on the formulation of Description Logic (DL) expressions that represent specific reasoning requests over the ontology. Each DL query is interpreted by the reasoner as a logical condition, and the output consists of the set of individuals that satisfy such condition, if any exist.

This mechanism allows the verification of both explicitly asserted knowledge and inferred relationships derived from the ontology axioms.

The overall reasoning workflow is illustrated in **Figure 3**, where an example query is shown. In this case, the input question is "Which soil has pH level of acidic and neutral?", formalized as the DL expression:

$$\text{SoilType} \sqcap (\text{hasPhLevel some Acidic}) \sqcap (\text{hasPhLevel some Neutral})$$

The inferred results returned by the reasoner are *Organic* and *Sandy*, confirming that the ontology correctly captures crop placement within cultivation areas.

³The inspiration for this image comes from the article [19]

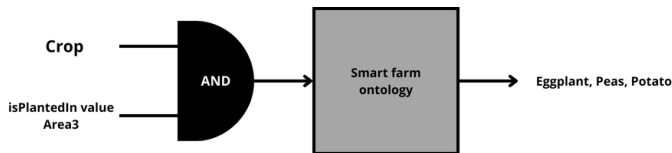


Figure 3: Logical process to test the ontology

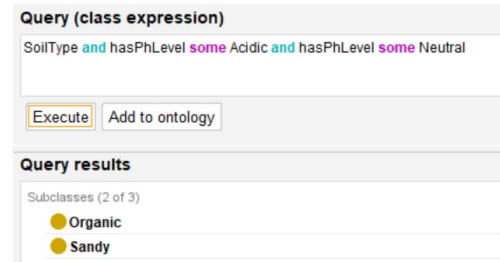


Figure 4: Protégé implementation to test the ontology

Different tests are performed to verify critical behavior such as disjoint class, inherited property:

- Query 1:** Which crop needs to be irrigated via DripLineDeployer and planted on Area 1?
DL: Crop and (needsTypeIrrigation some DripLineDeployer) and (isPlantedIn value Area1)
Results: Parsnip and Pepper.
- Query 2:** Which crop need to be irrigate via BoomSprayer and planted on Area 1?
DL: Crop and (needsTypeIrrigation some BoomSprayer) and (isPlantedIn value Area1) nothing
Results: owl:Nothing (it means there is no element that satisfy the DL).
- Query 3:** Which crops are planted in Area3?
DL: Crop and isPlantedIn value Area 3
Results: Eggplant, Peas and Potato.
- Query 4:** Which type of crops prefers to grow on Winter?
DL: Crop and seasonPreferred value Winter
Results: Celery, Parsnip and Squash.
- Query 5:** Which type of crops need to be planted with the transplanter?
DL: Crop needsTypeSeeding some Transplanter
Results: Broccoli, Cabbage, Cauliflower, Celery, Eggplant, Pepper and Tomato.
- Query 6:** Which tractors can plow?
DL: AutonomousTractor and hasTool some (performs some SoilPreparation)
Results: HighPowerVehicle

This subset of queries proves the validity of the model. In particular from **Query 6**, *hasTool* is the object property that links the power subclasses of the tractor to *Tool*, whereas the query asks the connection about *AutonomousTractor* and *AgriculturalOperation*.

The reasoner is therefore able to infer the relationship between tractors and agricultural operations by traversing the intermediate *Tool* concept. This result confirms that the ontology correctly supports chained reasoning over multiple object properties, validating its capability to recognize and infer concatenated relationships. Also, the **Pellet** reasoning is powerful enough to also understand concatenated queries.

To assess the practical viability of the ontology for real-time autonomous operations, It was measured **Pellet's** reasoning performance across 10 consecutive reasoner activations within a single Protégé session. Each measurement captured the time required to classify the ontology and execute the six DL queries. Recorded times were: 621 ms, 88 ms, 105 ms, 71 ms, 115 ms, 79 ms, 112 ms, 67 ms, 87 ms, and 107 ms. This measurements confirms that the ontology reasoning approach requires an overall average of **145.2 ms** for autonomous tractor decision-making.

6 Model Evaluation

This work presented an ontology-based model for representing an agricultural domain focused on autonomous tractor operations. The ontology captures key entities such as crops, soil types, fields, tools, agricultural operations, and sensors, together with their semantic relationships.

Through consistency checking and Description Logic queries, the model was validated to ensure logical coherence and correct reasoning behavior. The results demonstrate that the ontology is able to infer crop compatibility, select suitable agricultural operations and tools, and support field planning under soil and seasonal constraints.

Overall, the proposed ontology provides a structured and extensible foundation for autonomous decision-making in precision agriculture.

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